

dMAT-Style Mathematical Equations

Practice Set — no notes, no calculator, mental solving only

Low Difficulty

Question 1

$$A - 14 = 4$$

$$A - 14 = B$$

Question 2

$$14 - A = 6$$

$$A \div 4 = B$$

Question 3

$$3 + A = B$$

$$A - 5 = 6$$

Question 4

$$23 - A = 5$$

$$A - 12 = B$$

Question 5

$$A \div 5 = B$$

$$15 - A = 10$$

Medium Difficulty

Question 6

$$C \div 4 = A$$

$$6 + A = B$$

$$A + C = 5$$

Question 7

$$A - 2 = B$$

$$2 \times A + 2 \times C = 50$$

$$15 + C = A$$

Question 8

$$C + A = 4$$

$$C \div 3 = A$$

$$11 + A = B$$

Question 9

$$B \div 2 = A$$

$$12 - B = A$$

$$A \div 2 = C$$

Question 10

$$20 - B = A$$

$$A \div 4 = C$$

$$B \div 4 = A$$

High Difficulty (3 variables)**Question 11**

$$3 \times C = A$$

$$C \div 3 = B$$

$$C + A + B = 13$$

Question 12

$$B \div 2 = A$$

$$B + 2 \times C - A = 7$$

$$1 + B = C$$

Question 13

$$9 + A = B$$

$$4 \times A = C$$

$$C - A + B = 25$$

Question 14

$$A + C + B = 32$$

$$4 \times B = A$$

$$14 + B = C$$

Question 15

$$A + C + B = 14$$

$$10 + A = C$$

$$2 \times A = B$$

High Difficulty (4 variables)**Question 16**

$$B \div 4 = D$$

$$8 + B = A$$

$$2 \times B = C$$

$$B + A + C + 2 \times D = 26$$

Question 17

$$4 \times C = D$$

$$D + A + C + B = 25$$

$$C \div 3 = B$$

$$6 + C = A$$

Question 18

$$3 \times A = C$$

$$C + 2 \times D + B + A = 15$$

$$A + 1 = D$$

$$6 + A = B$$

Question 19

$$15 + B = C$$

$$B \div 3 = A$$

$$D + 2 \times B + C + A = 31$$

$$2 \times B = D$$

Question 20

$$A \div 3 = D$$

$$2 \times A = B$$

$$9 + A = C$$

$$D + C - 2 \times B + A = 4$$

Answer Key

Question 1	A = 18, B = 4
Question 2	A = 8, B = 2
Question 3	A = 11, B = 14
Question 4	A = 18, B = 6
Question 5	A = 5, B = 1
Question 6	A = 1, B = 7, C = 4
Question 7	A = 20, B = 18, C = 5
Question 8	A = 1, B = 12, C = 3
Question 9	A = 4, B = 8, C = 2
Question 10	A = 4, B = 16, C = 1
Question 11	A = 9, B = 1, C = 3
Question 12	A = 1, B = 2, C = 3
Question 13	A = 4, B = 13, C = 16
Question 14	A = 12, B = 3, C = 17
Question 15	A = 1, B = 2, C = 11
Question 16	A = 12, B = 4, C = 8, D = 1
Question 17	A = 9, B = 1, C = 3, D = 12
Question 18	A = 1, B = 7, C = 3, D = 2
Question 19	A = 1, B = 3, C = 18, D = 6
Question 20	A = 3, B = 6, C = 12, D = 1